IB Economics - Topic 1.1: Teacher and AI Grading Guide

What is economics? | SL/HL | Questions 1-20

Scope and assessment status

This original guide accompanies 20 complete Paper 1-style practice pairs. It is not official IB marking material. The official Topic 1.1 syllabus specifies explanation and diagram skills (AO2/AO4), rather than AO3 evaluation. The 15-mark parts here are therefore adapted evaluation practice using Topic 1.1 concepts, not a claim that these are official Topic 1.1 examination demands.

Quantity exception: the requested target is 50 pairs. This edition retains 20 pairs to limit substantive repetition and avoid importing the models and policies of later topics. Difficulty is scaled proportionally: 4 Easy, 10 Medium and 6 Hard. Every pair is shared SL/HL content. Difficulty denotes reasoning demand, not syllabus level.

Source basis

The IB Economics guide (first assessment 2022), printed pp. 22-23, governs scope. The uploaded SL and HL subject briefs confirm the course structure. Ellie Tragakes, Economics for the IB Diploma, second edition (2012), printed pp. 1-7 and 216-218, is used for conceptual checking only: its old syllabus numbering and assessment arrangements are not adopted. The six uploaded screenshots inform concise question phrasing and layout; no sample question is copied. AP materials are not used to expand IB scope.

Coverage blueprint

Scarcity, choice and resource use: 1-3, 5, 10, 13, 20

Basic questions and economic systems: 2, 4-6, 20

PPC assumptions, shapes, efficiency and growth: 2-3, 7-13

Circular flow and interdependence: 14-19

Social nature; microeconomic versus macroeconomic scale: 14, 19

Nine central concepts: Scarcity/choice: 1-3; efficiency: 6-9; equity: 5-6; well-being: 8-12, 19; sustainability: 3; change: 9-13; interdependence: 14-19; intervention: 4-6, 16

All question references below use Topic 1.1 and the official subtopic wording. Ranges in the blueprint denote overlapping coverage, not extra official subtopic numbers. No formal calculation is required.

How to mark and use this guide

Apply a best-fit judgment separately to part (a), out of 10, and part (b), out of 15. Add them for a score out of 25. Indicative content describes sound routes through each question; it is not a marks-per-keyword checklist. Do not demand every suggested point or example.

Part (a): explanatory quality

Use the 0, 1-2, 3-4, 5-6, 7-8 and 9-10 bands as a progression from no relevant achievement, through limited and descriptive responses, to partially developed, clear and fully developed explanations. Judge relevance, correct terminology, causal reasoning and the explanation of any appropriate diagram together. The highest band requires the full task to be addressed; mere definitions and lists are insufficient.

Part (b): analytical and evaluative quality

Use the 0, 1-3, 4-6, 7-9, 10-12 and 13-15 bands as a progression from no relevant achievement, through weak and superficial responses, to developed but unbalanced analysis, balanced evaluation and sustained, well-supported judgment. Assess application of real examples, accurate theory, diagram use where appropriate and responsiveness to the proposition. These are abridged practice descriptors, not reproduced official markbands.

Examples and alternative answers

The suggested cases are optional starting points. Relevant, accurate alternatives receive equal credit. A country name alone is not developed application: the response must identify the relevant circumstances and explain the mechanism. No exact statistic is required. Hypothetical illustrations can clarify theory but do not substitute for a real-world example when the question asks for one. The conclusions below illustrate defensible judgments, not prescribed opinions.

Diagram assessment for a multimodal grader

Check axis meaning, curve shape or arrow direction, labels, starting position, change and economic interpretation against the question-specific specification. Accept reversed PPC axes, different letter labels and equivalent circular-flow layouts when consistent. Curves need not be artistic or drawn to scale. PPCs and circular flows do not require market-equilibrium intersections or welfare areas. If a feature is illegible or ambiguous, flag it for review; do not assume that it is correct. Do not deduct the same conceptual error repeatedly as unrelated omissions.

Record for each response

Record the two marks, the evidence supporting the band choices, the main conceptual strengths and gaps, any diagram interpretation uncertainty, and one actionable improvement for each part. Give credit for valid reasoning outside the indicative route; do not reward unrelated advanced theory in place of answering the question.

Difficulty rationale

Easy pairs involve direct resource constraints or simple interdependence. Medium pairs require linked mechanisms and conditional evaluation. Hard pairs require distinguishing apparently similar outcomes, resolving intertemporal choices or integrating competing explanations within Topic 1.1.

Question 1

Easy | SL/HL | 25 marks

Topic 1.1: The problem of choice: scarcity; factors of production; opportunity cost

(a) [10] Explain why an increase in an economy's resources does not necessarily remove the problem of scarcity.

(b) [15] Using real-world examples, discuss the view that technological progress can eliminate the need for economic choices.

Part (a) - Indicative answer and marking guidance

Definitions. Scarcity is the insufficiency of resources relative to needs and wants. Factors of production are land, labour, capital and entrepreneurship. Opportunity cost is the next best alternative forgone.

Reasoning. More or better resources increase what can be produced, but do not establish that every competing want can be met. Explain how resources still have alternative uses and why producing additional output can require sacrificing something else. Distinguish a reduction in a particular constraint from the disappearance of the economic problem.

Diagram. No diagram required. If a PPC is used, label the axes as quantities of two outputs, show PPC1 and an outward PPC2, and retain unattainable combinations beyond PPC2. Explain the remaining choice along PPC2; a shift alone is insufficient.

Common errors. Equating scarcity with poverty; treating capital as money; asserting that every individual want must always grow; claiming that a larger resource stock makes all combinations attainable.

Alternatives. Credit discussion of changing wants, finite time, or complementary resource constraints. A response can explain continuing scarcity without assuming that all wants increase.

Part (b) - Indicative answer and marking guidance

Main argument. Technology can expand feasible output and reduce resource requirements per unit; innovations can overcome particular shortages. This can widen feasible choices and improve access to goods.

Counterargument and conditions. Research, equipment, energy and skilled labour themselves use resources. A new technology may relieve one constraint while leaving others. Greater productive capacity does not determine the desired output mix or its distribution.

Real-world application. S1: Singapore's recycled-water technology expands usable water supply but requires treatment plants and other inputs. S6: Korea's investment in technology and skills expanded capacity. Apply either case to the remaining resource choices; do not just name it.

High-quality evaluation and judgment. A strong answer separates overcoming a specific constraint from removing scarcity in general. A defensible judgment is that technology changes the available choices rather than eliminating the need to choose.

Question 2

Medium | SL/HL | 25 marks

Topic 1.1: The problem of choice: opportunity cost; PPC: choice

(a) [10] Using a production possibilities curve, explain the opportunity cost of reallocating resources from education to health care.

(b) [15] Using real-world examples, evaluate the view that governments should base resource allocation decisions only on financial costs.

Part (a) - Indicative answer and marking guidance

Definitions. Opportunity cost is the value of the next best alternative forgone. Financial cost is a monetary payment; it need not capture every forgone use of resources.

Reasoning. Assume productive efficiency and fixed resources and technology. Additional health services require resources previously used in education. Explain the forgone educational output and distinguish it from the money paid to health providers. Trace the reallocation rather than merely stating a definition.

Diagram. Required PPC: horizontal axis quantity of health services, vertical axis quantity of education services. On one downward-sloping PPC, A has less health care and more education; B has more health care and less education. Mark the health gain and education forgone with projections. No price equilibrium or welfare area is required. Reversed axes with consistent movements are valid.

Common errors. Shifting the PPC merely because priorities change; calling the health gain the opportunity cost; adding every possible alternative instead of identifying the next best alternative.

Alternatives. A straight or bowed PPC is acceptable if the stated opportunity-cost assumption is consistent. Credit recognition that idle suitable resources could permit some extra output without the depicted sacrifice.

Part (b) - Indicative answer and marking guidance

Main argument. Financial costs help governments respect budgets, compare procurement options and identify avoidable waste. They are observable and important for implementation.

Counterargument and conditions. The next best public service forgone, scarce staff time and future resource consequences may differ from financial outlays. A low-cost project may meet less urgent needs or exclude disadvantaged groups.

Real-world application. S2: the NHS commitment to effective and fair use of finite resources supports comparison of health choices. A student may also use a well-explained local school or hospital allocation decision. Do not require numerical cost-benefit analysis.

High-quality evaluation and judgment. High-quality evaluation compares financial affordability with economic sacrifice and social priorities. The conclusion should explain when monetary costs are informative and why they are insufficient on their own.

Question 3

Hard | SL/HL | 25 marks

Topic 1.1: The problem of choice: scarcity and sustainability; PPC: future possibilities

(a) [10] Explain two ways in which the use of natural resources in current production can affect an economy's future production possibilities.

(b) [15] Using real-world examples, discuss whether meeting present needs should take priority over preserving resources for future generations.

Part (a) - Indicative answer and marking guidance

Definitions. Land includes natural resources. Sustainability concerns meeting present needs without undermining future generations' ability to meet theirs. Production possibilities describe feasible output with available resources and technology.

Reasoning. Develop two distinct mechanisms. Depleting finite resources or damaging renewable-resource quality can reduce future productive capacity. Alternatively, current resource use can support equipment, skills or resource-restoring activity that expands future capacity. Explain the conditions for each, not an automatic offset.

Diagram. No diagram required by the stem. An appropriate supplementary PPC uses two output quantities, with PPC1 and a future inward or outward PPC2. Identify the resource or technology change causing each shift. Do not treat a reallocation along a frontier as evidence of depletion.

Common errors. Treating every natural resource as non-renewable; assuming all current production destroys future capacity; claiming an outward PPC proves environmental sustainability.

Alternatives. Credit regeneration exceeding extraction, conservation improving resource quality, or present investment in less resource-intensive methods. Two independently developed mechanisms are needed for a strong explanation.

Part (b) - Indicative answer and marking guidance

Main argument. Urgent present needs for food, housing and basic services can justify resource use. Present production can also finance or create assets that benefit future generations.

Counterargument and conditions. Irreversible depletion restricts future choices. Future people cannot participate directly in current allocation decisions. The balance depends on urgency, renewability, regeneration and whether current gains create durable alternatives.

Real-world application. S1: Singapore's water-reuse infrastructure illustrates meeting current needs while reducing a resource constraint. S5: Japan's reconstruction illustrates pressing present needs and future capacity. These cases support reasoning; neither proves all resource use is sustainable.

High-quality evaluation and judgment. A strong response identifies whose needs are prioritised, distinguishes essential consumption from less urgent wants, and gives a conditional judgment about preserving future options. Formal externality theory or discounting is not required.

Question 4

Medium | SL/HL | 25 marks

Topic 1.1: The problem of choice: basic economic questions; economic systems

(a) [10] Explain how a market economy and a planned economy differ in deciding what to produce and how to produce.

(b) [15] Using real-world examples, evaluate the view that market allocation is always preferable to government planning.

Part (a) - Indicative answer and marking guidance

Definitions. Market allocation relies chiefly on decentralised choices and exchange. Planning relies chiefly on government allocation decisions. What concerns output composition and quantities; how concerns production methods and input combinations.

Reasoning. Explain how consumer purchases and producer decisions influence the output mix in a market system, and how producers select methods within available resources. Contrast government production priorities and instructions in a planned system. Connect each mechanism to both basic questions.

Diagram. No diagram required. A demand-supply equilibrium diagram is not needed for full credit and should not become the assessment target.

Common errors. Describing mixed economies as purely planned; asserting that markets have no rules; discussing only for whom when the stem asks what and how.

Alternatives. Credit well-explained differences in public ownership, private ownership, incentives and information where these illuminate the two decisions. Ideal systems can be compared without naming a completely pure real economy.

Part (b) - Indicative answer and marking guidance

Main argument. Decentralised choices can use local information and respond to varied preferences. Incentives can encourage producers to economise on scarce inputs and adapt output.

Counterargument and conditions. Ability to pay may not reflect urgency of need. Coordinated allocation can prioritise basic services or long-term needs, but planning may face information and responsiveness problems. Compare mechanisms, not absolute labels.

Real-world application. S3: Vietnam's transition towards a more market-oriented system supports discussion of decentralised production choices. S2: NHS provision shows a continuing role for public allocation within a mixed economy. Neither country represents a pure system.

High-quality evaluation and judgment. A strong judgment rejects or defends the universal claim by specifying the goods, objectives and institutional conditions. No market-failure diagram, tax analysis or elasticity knowledge is required.

Question 5

Easy | SL/HL | 25 marks

Topic 1.1: The problem of choice: free goods; for whom to produce; economic systems

(a) [10] Explain how scarcity affects the allocation of health care when patients are not charged at the point of use.

(b) [15] Using real-world examples, discuss whether access to essential goods and services should depend on consumers' ability to pay.

Part (a) - Indicative answer and marking guidance

Definitions. An economic good uses scarce resources and has an opportunity cost. A free good is available without sacrificing an alternative use in the relevant context. A zero price to a user does not by itself make a good free in the economic sense.

Reasoning. Health care uses scarce staff, equipment, buildings and time despite a zero user charge. Explain both the allocation of resources between health care and other uses and allocation among patients through need, waiting time or other criteria. Distinguish the user's payment from society's resource cost.

Diagram. No diagram required. A PPC may support the alternative-use argument but is not necessary. If used, axes must be quantities of health care and another output, with a clear sacrifice between efficient combinations.

Common errors. Calling all publicly provided services free goods; assuming no charge means unlimited access; treating the government budget as the only scarce resource.

Alternatives. Credit alternative non-price allocation mechanisms with developed resource reasoning. Do not require health-care-specific institutional knowledge.

Part (b) - Indicative answer and marking guidance

Main argument. Payment-based access allows consumers to express preferences and can provide resources and incentives for supply. It may reduce some uses that consumers value less than the price paid.

Counterargument and conditions. Income and wealth differ, so ability to pay may exclude people with urgent needs. Need-based provision can widen access but still requires prioritisation and funding; waiting time or limited choice can remain.

Real-world application. S2: the NHS principle of access based on clinical need rather than ability to pay is directly relevant. S1 can illustrate another essential resource with substantial production costs. Examples must connect access criteria to outcomes.

High-quality evaluation and judgment. A strong response weighs resource constraints, urgency of need, consumer choice and access. A mixed allocation arrangement is a valid conclusion if its trade-offs are explained.

Question 6

Medium | SL/HL | 25 marks

Topic 1.1: The problem of choice: mixed economy; efficiency; equity; intervention

(a) [10] Explain how private producers and the government can both contribute to resource allocation in a mixed economy.

(b) [15] Using real-world examples, evaluate the view that increasing government ownership necessarily improves both efficiency and equity.

Part (a) - Indicative answer and marking guidance

Definitions. A mixed economy combines private economic activity with government allocation and intervention. Efficiency concerns avoiding waste and using resources effectively; equity concerns fairness and need not mean identical outcomes.

Reasoning. Develop distinct private and public mechanisms. Private producers respond to purchasers and organise production; government can provide services, own productive assets and set allocation priorities. Explain their coexistence and interaction rather than offering a list of sectors.

Diagram. No diagram required. A PPC can show productive efficiency but cannot itself establish equitable distribution. Do not require technical allocative-efficiency conditions.

Common errors. Equating all intervention with ownership; treating equity as a point on the PPC; assuming every public enterprise is inefficient or every private enterprise efficient.

Alternatives. Credit contracted private production of publicly funded services and state-owned firms operating alongside private firms, with clear identification of ownership versus funding.

Part (b) - Indicative answer and marking guidance

Main argument. Public ownership can direct resources towards access and continuity of essential services, and enable coordinated investment. These features may support fairness and reduce some forms of waste.

Counterargument and conditions. Outcomes depend on management, information, incentives and accountability. Ownership alone does not determine who benefits or whether inputs are wasted. Public ownership also uses resources that could serve other priorities.

Real-world application. S1: PUB's water system provides a concrete public allocation case. S2: NHS funding and provision illustrate that public allocation can coexist with varied organisational arrangements. Use S3 as an alternative mixed-system context.

High-quality evaluation and judgment. High-quality work assesses efficiency and equity separately, then considers whether both improve under the stated conditions. A judgment should address necessarily, not merely list advantages and disadvantages of government.

Question 7

Hard | SL/HL | 25 marks

Topic 1.1: PPC: assumptions; increasing versus constant opportunity cost; choice

(a) [10] Using production possibilities curves, explain why opportunity cost may be constant in one economy but increasing in another.

(b) [15] Using real-world examples, evaluate the usefulness of the production possibilities curve for deciding what an economy should produce.

Part (a) - Indicative answer and marking guidance

Definitions. A PPC shows maximum feasible combinations of two outputs with given resources and technology. Constant opportunity cost means a constant sacrifice per additional unit; increasing opportunity cost means that sacrifice rises as production is reallocated.

Reasoning. With constant relative suitability of resources, successive equal increases in one output sacrifice equal amounts of the other. With heterogeneous resources, inputs increasingly unsuited to the expanding activity must be transferred, so the sacrifice rises. Keep technology and resources fixed within each diagram.

Diagram. Required: two PPC panels with quantity of good X horizontally and quantity of good Y vertically. One frontier is straight; the other is bowed outward from the origin. Mark successive equal horizontal gains: equal vertical sacrifices on the straight line, increasing vertical sacrifices on the bowed line. Labels and explanations must be consistent; no equilibrium or shaded welfare area is needed.

Common errors. Using a frontier bowed towards the origin for increasing opportunity cost; comparing unequal output increments without explaining per-unit costs; attributing movement along the PPC to technological progress.

Alternatives. Credit numerical illustrations used to clarify, not replace, causal reasoning. Both diagrams may describe different production conditions rather than literal whole-country estimates.

Part (b) - Indicative answer and marking guidance

Main argument. The PPC clarifies feasible choices, resource constraints and sacrifices, and distinguishes attainable from unattainable plans. It can make the trade-off between public services or current and future output explicit.

Counterargument and conditions. The frontier does not reveal preferences, distribution or the social priority of different outputs. Aggregating many goods into two categories conceals detail; an estimated frontier may be uncertain or change over time.

Real-world application. S2 supports a health-service allocation application; S6 supports current investment and future possibilities. Treat any drawn national PPC as an analytical illustration, not an empirically measured frontier supplied by these sources.

High-quality evaluation and judgment. A strong judgment distinguishes useful information about feasibility from sufficient information to select a socially preferred combination. The evaluation must concern the PPC specifically, without requiring Topic 1.2 methodology.

Question 8

Easy | SL/HL | 25 marks

Topic 1.1: PPC: unemployment of resources; productive efficiency; actual growth

(a) [10] Using a production possibilities curve, explain how an economy can increase the production of both goods without an increase in its resources.

(b) [15] Using real-world examples, discuss the view that bringing unemployed resources into production always improves economic well-being.

Part (a) - Indicative answer and marking guidance

Definitions. Productive efficiency means that more of one output cannot be produced without less of another, given resources and technology. Unemployed resources are available inputs not currently used. Economic well-being includes material living conditions and broader security and quality of life.

Reasoning. Begin inside the PPC with unemployment or inefficient use of resources. Employ suitable unused inputs or reduce waste so that both outputs rise towards the existing frontier. Explain why no expansion of the frontier is needed and why the same gain is not feasible from every efficient starting point.

Diagram. Required PPC: two output quantities on the axes; unchanged frontier. A lies strictly inside it, and B lies on or closer to it, northeast of A, so both quantities rise. Distinguish the outward movement of the production point from an outward shift of the PPC. Reversed axes and a B still inside are acceptable.

Common errors. Shifting the frontier solely because idle inputs are used; starting on the PPC and increasing both outputs with unchanged capacity; treating all non-employment, including chosen leisure, as identical.

Alternatives. Credit better organisation or repair of usable idle equipment if this improves utilisation rather than increases the assumed resource stock. The explanation must establish a feasible starting point.

Part (b) - Indicative answer and marking guidance

Main argument. Using involuntarily idle resources can raise output and earned income, improve access to goods and reduce insecurity. More efficient use can produce gains without sacrificing the other depicted output.

Counterargument and conditions. Well-being depends on the output produced, who receives it and the conditions of production. Some apparent non-use protects future resources or reflects valued leisure. Harmful or unwanted output need not improve well-being.

Real-world application. S4: pandemic restrictions and reduced spending provide a context for idle resources and recovery; acknowledge that some work was constrained by health risks. S5 supports discussion of restoring usable facilities after disruption.

High-quality evaluation and judgment. The best answers distinguish involuntary idleness from all forms of non-use and reject an automatic equivalence between more production and better lives. A positive but conditional judgment is valid.

Question 9

Medium | SL/HL | 25 marks

Topic 1.1: PPC: actual growth; growth in production possibilities; well-being

(a) [10] Using production possibilities curves, explain the difference between actual economic growth and growth in production possibilities.

(b) [15] Using real-world examples, evaluate the view that an outward shift of an economy's production possibilities curve guarantees an improvement in economic well-being.

Part (a) - Indicative answer and marking guidance

Definitions. Actual growth is an increase in real output. Growth in production possibilities is an increase in productive capacity. Economic well-being is broader than the maximum quantity of output that could be produced.

Reasoning. Show actual growth through greater utilisation of given capacity and potential growth through more or better resources or technology. Explain that actual output can also rise as capacity expands; the categories are distinct, not mutually exclusive. More potential output need not be realised immediately.

Diagram. Required: quantity of output X horizontally and output Y vertically. Panel 1 has A inside PPC1 and B northeast of A nearer the same PPC. Panel 2 shows outward PPC2 beyond PPC1 from improved resources or technology. Label actual production separately if shown; do not imply the economy automatically reaches PPC2.

Common errors. Calling every output increase a frontier shift; defining capacity growth as a move along the frontier; assuming actual and potential growth must occur separately.

Alternatives. Accept one clearly labelled composite diagram instead of two panels. Accept capacity growth enabling actual growth if the distinction between actual output and feasible capacity remains explicit.

Part (b) - Indicative answer and marking guidance

Main argument. Expanded capacity permits more goods and services and may relax constraints on access to essentials. It creates opportunities for improved material living conditions.

Counterargument and conditions. Unused new capacity does not automatically increase consumption. Distribution, the output mix, working conditions and depletion of resources can change how people benefit. A two-good frontier does not measure these outcomes.

Real-world application. S6: Korea's sustained investment in education, infrastructure and technology illustrates expanding possibilities. S1 illustrates improved water capacity; link that capacity to realised access rather than assuming an automatic benefit to everyone.

High-quality evaluation and judgment. A strong conclusion treats an outward shift as an opportunity, then identifies the conditions under which it becomes improved well-being. Detailed GDP measurement, development indices or AD-AS analysis are unnecessary.

Question 10

Hard | SL/HL | 25 marks

Topic 1.1: PPC: choice over time; capital goods; opportunity cost

(a) [10] Using a production possibilities curve, explain how producing more capital goods can affect the production of consumer goods now and in the future.

(b) [15] Using real-world examples, discuss whether an economy should prioritise capital goods over consumer goods.

Part (a) - Indicative answer and marking guidance

Definitions. Capital goods are produced inputs used to produce other goods and services. Consumer goods satisfy needs and wants directly. Opportunity cost refers to the next best output forgone.

Reasoning. At productive efficiency, increasing capital-goods production along the current PPC sacrifices consumer output. Useful additional capital can then expand future capacity, permitting more consumer output. The link depends on use, suitability and maintenance; it is not a guaranteed shift of a particular size.

Diagram. Required: horizontal axis consumer goods, vertical axis capital goods. A and B lie on PPC1; B has more capital goods and fewer consumer goods. Show a later outward PPC2 and identify investment as the cause. Avoid treating the movement to B as the frontier shift itself. Reverse axes are acceptable with consistent changes.

Common errors. Calling a share purchase production of capital goods; claiming all investment has immediate benefits without sacrifice; assuming every machine is useful or that capital never depreciates.

Alternatives. Credit a comparison of two future paths if the difference in present allocation is explained. No growth-model equations or numerical investment calculation is required.

Part (b) - Indicative answer and marking guidance

Main argument. More useful capital can expand future choices, improve productive capacity and support higher future consumption. Deferring some consumption may therefore benefit future households.

Counterargument and conditions. Current essential needs may be urgent; some groups bear the sacrifice while others receive later benefits. Poorly chosen or unused capital may add little capacity. Complementary skills and resources affect outcomes.

Real-world application. S6: Korea's investment-led transformation supports the future-capacity argument; distinguish physical capital from education improving labour quality. S2 offers a context for immediate health needs competing for resources.

High-quality evaluation and judgment. A strong judgment weighs the scale and distribution of present sacrifice against credible future gains, taking the economy's starting conditions into account. A universal capital-first rule is not required or presumed.

Question 11

Medium | SL/HL | 25 marks

Topic 1.1: PPC: changes in technology; changes in production possibilities

(a) [10] Using a production possibilities curve, explain how a technological improvement affecting only one good can change an economy's production possibilities.

(b) [15] Using real-world examples, evaluate the view that technological improvements in production benefit all groups in an economy.

Part (a) - Indicative answer and marking guidance

Definitions. Technology concerns methods of production. A sector-specific improvement changes what can be produced from given resources in that activity. A gain in potential production is not identical to a gain for every stakeholder.

Reasoning. Explain how the same resources can produce more of the affected good while the maximum possible output of the unaffected good remains unchanged under the stated simplification. At intermediate allocations, improved productivity expands feasible combinations. Do not assume the actual output mix is predetermined.

Diagram. Required: horizontal quantity of technologically improved good X; vertical quantity of other good Y. PPC2 extends the X-intercept relative to PPC1 while sharing the Y-intercept. Draw PPC2 outside PPC1 for combinations where the improvement is usable. Label both curves. A smooth pivot-like expansion is acceptable; do not require a mechanically rotated identical curve.

Common errors. Requiring both intercepts to increase under an explicitly sector-specific change; labelling price on a PPC axis; claiming a frontier shift shows every group's income rising.

Alternatives. Credit spillovers in part (b), but the diagram in part (a) should respect the one-good assumption. Alternative axes are valid.

Part (b) - Indicative answer and marking guidance

Main argument. Improved methods can increase available output and reduce input requirements, potentially widening consumption opportunities. Some workers and resource owners can gain opportunities in the expanding activity.

Counterargument and conditions. Benefits depend on access to the technology and the goods produced. Workers whose skills are less useful and owners of displaced equipment may lose opportunities. New input requirements and resource use can limit broad gains.

Real-world application. S1: NEWater illustrates a specific improvement in usable-water production. S6 provides a broader technological change context. The source establishes the change, while distributional outcomes must be argued rather than asserted as documented facts.

High-quality evaluation and judgment. High-quality evaluation distinguishes aggregate productive potential from distribution among consumers, workers and owners. Conclusions should identify conditions for widespread benefit without requiring labour-market or market-power models.

Question 12

Hard | SL/HL | 25 marks

Topic 1.1: PPC: resource destruction; unemployment; actual versus potential output

(a) [10] Using production possibilities curves, explain the different effects of the destruction of capital and the temporary non-use of undamaged capital.

(b) [15] Using real-world examples, evaluate the view that restoring an economy's previous level of output is a sufficient measure of recovery after a natural disaster.

Part (a) - Indicative answer and marking guidance

Definitions. Physical capital is produced equipment and structures used in production. Productive capacity depends on usable resources and technology; actual output also depends on their utilisation.

Reasoning. Destroyed machinery or infrastructure reduces usable productive resources and can move the PPC inward. Temporary non-use of intact, accessible equipment can reduce actual output within an unchanged frontier. Explain how repairs or reconstruction differ from restarting idle resources; a real disaster can combine both.

Diagram. Required: two output quantities on each panel. Panel 1: PPC2 lies within PPC1 because relevant capital is destroyed. Panel 2: actual production moves from A towards an interior B with a constant PPC as intact capital becomes idle. The fall need not be identical in both goods. Clearly distinguish the cases; no equilibrium or welfare areas apply.

Common errors. Shifting capacity inward for every temporary fall in use; treating destroyed capital as immediately available; assuming actual output necessarily equals the post-disaster frontier.

Alternatives. Credit a sector-specific inward change for damage concentrated in one activity, or one well-labelled combined diagram that keeps the mechanisms separate.

Part (b) - Indicative answer and marking guidance

Main argument. Returning to previous output can indicate restored production, employment opportunities and access to goods. It is a relevant benchmark for the scale of recovery.

Counterargument and conditions. Reconstruction output may replace destroyed assets rather than add to well-being. Housing, access to essentials and distribution may remain impaired even when total output recovers. The composition and resilience of rebuilt capacity matter.

Real-world application. S5: the 2011 Great East Japan Earthquake and reconstruction provide the core case. Distinguish the report's documented disruption and rebuilding from the student's analytical judgment about welfare and capacity.

High-quality evaluation and judgment. A strong response treats output restoration as useful evidence rather than a complete criterion. Judgments should consider the recovered output mix, people affected and productive assets, with no requirement for national-accounting calculation.

Question 13

Medium | SL/HL | 25 marks

Topic 1.1: The problem of choice: labour and capital; PPC: resource quantity and quality

(a) [10] Explain how an increase in the quantity of labour and an improvement in the quality of labour can each expand an economy's production possibilities.

(b) [15] Using real-world examples, discuss whether improving labour skills is more important than increasing the size of the labour force for expanding production possibilities.

Part (a) - Indicative answer and marking guidance

Definitions. Labour is human physical and mental effort in production. Labour quantity concerns available workers or working time; labour quality concerns relevant skills, health and capabilities. Physical capital is distinct from human skills.

Reasoning. More available workers or working time can permit greater output where complementary resources exist. Improved skills can raise what existing workers can produce and enable different production methods. Explain each chain through resource use to productive capacity rather than to demand for output.

Diagram. No diagram required by the stem. A supporting PPC has quantities of two outputs on the axes and an outward PPC2, with the specific labour change labelled. An asymmetric expansion is valid if skill gains affect one activity more strongly.

Common errors. Assuming a larger population automatically means more usable labour; treating education spending as an immediate equal gain in all skills; treating skills as additional physical machines.

Alternatives. Credit health, relevant experience or training as quality improvements. Accept conditional expansion where land, capital or suitable production opportunities are limited.

Part (b) - Indicative answer and marking guidance

Main argument. Skills can make existing inputs more productive and support more complex production. In an economy with limited workers, improved quality can expand possibilities without a large labour-force increase.

Counterargument and conditions. Where labour availability is the binding constraint, more workers may add more capacity than further training. Skills take time and resources to develop and must fit available capital and production needs. The two changes can be complementary.

Real-world application. S6: Korea's sustained emphasis on education and technology supports a skills application. S5 can illustrate that skilled labour still needs functioning infrastructure after disruption. Do not present a precise quantitative comparison without evidence.

High-quality evaluation and judgment. A high-quality judgment identifies which constraint is binding, the relevant timescale and complementarities. A conditional answer is stronger than claiming that either more people or more training is always superior.

Question 14

Easy | SL/HL | 25 marks

Topic 1.1: Modelling the economy: circular flow; household-firm interdependence

(a) [10] Using a circular flow of income diagram, explain the interdependence between households and firms in a closed economy without government.

(b) [15] Using real-world examples, discuss the view that households depend more on firms than firms depend on households.

Part (a) - Indicative answer and marking guidance

Definitions. Households supply factors of production and purchase output. Firms organise production, purchase factor services and sell output. Interdependence means each group's activity relies on the other.

Reasoning. Trace real factor services from households to firms, factor income back to households, goods and services from firms to households, and consumption expenditure back to firms. Explain why factor payments support spending and spending supports firms' revenue. State the two-sector simplification.

Diagram. Required: households and firms, optionally factor and product markets. Factors: households to firms; factor income: firms to households; output: firms to households; consumption expenditure: households to firms. Label money and real flows distinctly. No S, T, M, I, G or X is needed in this closed, no-government simplification; no numerical equilibrium point or area applies.

Common errors. Sending factor income in the same direction as factor services; treating households only as buyers; adding government or foreign sectors as essential features of the stated model.

Alternatives. Credit circular or box-and-arrow layouts with equivalent correct directions. Household ownership of firms is acceptable if it clarifies, rather than obscures, the two roles.

Part (b) - Indicative answer and marking guidance

Main argument. Households rely on firms for many goods, services and employment incomes. A disruption to production can therefore affect access and household income.

Counterargument and conditions. Firms rely on household labour, other factor services and consumer spending. Households may have subsistence, self-employment or public-service alternatives, while firms can have other purchasers in the real economy. Relative dependence varies by context and time.

Real-world application. S4: reduced household spending during the pandemic illustrates how firms can be affected by household choices. S5: disrupted production in Japan illustrates the reverse link. Explain the relationship instead of merely citing an event.

High-quality evaluation and judgment. A strong judgment distinguishes mutual dependence from a measurable claim that one side is always more dependent. The student may extend beyond the simple model in evaluation while retaining the household-firm focus.

Question 15

Medium | SL/HL | 25 marks

Topic 1.1: Modelling the economy: financial sector; saving and investment

(a) [10] Using a circular flow of income diagram, explain how saving and investment connect households, firms and the financial sector.

(b) [15] Using real-world examples, evaluate the view that an increase in household saving is always beneficial for an economy.

Part (a) - Indicative answer and marking guidance

Definitions. Saving is income not spent on current consumption. Investment is expenditure on produced capital goods. The financial sector connects financing and spending decisions; saving and investment are different economic activities.

Reasoning. Show saving as a withdrawal from current consumption expenditure and investment as an addition to spending on domestically produced capital goods. Explain that financial institutions can help channel finance to productive expenditure, but additional saving does not automatically produce matching additional investment.

Diagram. Required circular-flow diagram: households, firms and a financial-sector box or equivalent. Label S leaving the main spending-income loop towards the financial sector and I returning towards firms' production or the expenditure loop. Retain consumption and factor-income directions. Distinguish the conceptual spending flows from a literal account of bank money creation. No quantitative equilibrium is required.

Common errors. Calling the purchase of an existing share new physical investment; treating saving as money destroyed; claiming banks can lend only pre-existing household deposits; assuming S and planned I rise identically at every moment.

Alternatives. Accept direct financial-market links instead of a bank box. A correct discussion of investment expectations is relevant without loanable-funds or money-market diagrams.

Part (b) - Indicative answer and marking guidance

Main argument. Saving can provide households with security and contribute to financing future productive investment. If matched by useful investment, it can support future capacity and consumption.

Counterargument and conditions. Reduced consumption can lower firms' sales and household income if offsetting expenditure does not rise. Saving may be held rather than finance new production. Effects vary across households and between present and future.

Real-world application. S4: the Bank of England's 2020 household survey documents increased saving alongside reduced spending, unevenly across households. S6 illustrates the potential importance of productive investment, without proving a one-for-one causal link from saving.

High-quality evaluation and judgment. High-quality evaluation separates private financial security, current circular-flow effects and possible future capacity gains. It should state the conditions connecting saving with useful investment. A multiplier calculation is unnecessary.

Question 16

Medium | SL/HL | 25 marks

Topic 1.1: Modelling the economy: government sector; taxation and government spending

(a) [10] Using a circular flow of income diagram, explain the different roles of taxation and government spending.

(b) [15] Using real-world examples, discuss whether an increase in government spending necessarily increases the size of the circular flow of income.

Part (a) - Indicative answer and marking guidance

Definitions. Taxation is a leakage from the private spending-income loop. Government purchases of goods and services are an injection. Transfers are not identical to direct purchases and should not automatically be labelled G.

Reasoning. Explain how taxation diverts part of income away from private consumption and how public purchases create revenue for producers and factor incomes. The government may both withdraw and inject funds; the net effect depends on other flows, not just the existence of a government sector.

Diagram. Required: households, firms, government and the main consumption/factor-income loop. T is directed towards government from the income loop; G returns from government to purchases of current output. Alternative layouts may include firm taxes or direct public factor payments if explained consistently. No need for exact values, balanced budget or equilibrium point.

Common errors. Drawing taxes as an injection; treating every transfer as direct G; assuming government purchases add to spending with no possible offset; claiming T must equal G each period.

Alternatives. Accept a fuller five-sector diagram if the government flows remain clear. Credit an explanation that a transfer can affect subsequent consumption, while distinguishing it from direct government purchases.

Part (b) - Indicative answer and marking guidance

Main argument. Additional purchases of domestic output can raise firms' receipts and earned incomes. With usable idle resources and limited offsetting withdrawals, the income flow can expand.

Counterargument and conditions. Higher taxes or other reductions in private spending can offset the purchase increase. Purchases of imported output leak abroad. Where resources are fully used, financial spending need not imply a matching increase in real output. No advanced policy model is necessary.

Real-world application. S5: Japan's reconstruction spending supports an application of government purchases. S4 documents pandemic income support; distinguish transfers from direct purchases rather than misclassifying all support as G.

High-quality evaluation and judgment. A strong judgment compares the change in injections with relevant leakages and resource availability. Assess reasoning about the overall flow, not the mere assertion that more G is expansionary.

Question 17

Medium | SL/HL | 25 marks

Topic 1.1: Modelling the economy: foreign sector; imports and exports

(a) [10] Using a circular flow of income diagram, explain why expenditure on imports is a leakage and expenditure on exports is an injection.

(b) [15] Using real-world examples, evaluate the view that an increase in export receipts necessarily raises domestic income.

Part (a) - Indicative answer and marking guidance

Definitions. Imports are purchases of foreign-produced goods and services; exports are domestic output purchased by foreign residents. The relevant boundary is the domestic spending-income loop, not whether the activity has any value.

Reasoning. Import expenditure directs part of domestic spending to foreign producers rather than domestic output. Foreign spending on exports adds to domestic producers' sales and can support factor incomes. Explain the direction of the money flows, distinguished from the movement of goods.

Diagram. Required circular-flow diagram with domestic households/firms and a foreign sector. M leaves the domestic expenditure loop; X enters towards domestic output purchases or firms' receipts. Label money directions clearly. Goods move in the opposite directions if drawn. Taxes and saving may be omitted; no exchange-rate graph or equilibrium is needed.

Common errors. Calling imports a loss with no goods received; reversing goods and payment directions; confusing export sales receipts with the income retained domestically after imported inputs.

Alternatives. Credit imports of capital or intermediate goods as still a current expenditure leakage with possible productive benefits. Accept a full five-sector diagram with the foreign links highlighted.

Part (b) - Indicative answer and marking guidance

Main argument. Higher export receipts add spending on domestic production and can create incomes for domestic workers and other resource owners. The link is stronger where production has substantial domestic input content.

Counterargument and conditions. Imported inputs and other withdrawals can absorb part of the receipts. Other domestic spending or injections may fall at the same time. Capacity constraints can limit real output gains. A rise in gross receipts therefore does not establish the size of a net domestic-income change.

Real-world application. S7: the World Bank's Vietnam report links manufacturing and jobs to exports while highlighting the need for stronger domestic linkages. Apply this to domestic input content; do not infer that gross export receipts equal domestic income.

High-quality evaluation and judgment. A strong response distinguishes an injection's initial direction from its net effect on the wider flow. Comparative advantage, exchange-rate determination, trade elasticities and balance-of-payments policy are outside the required response.

Question 18

Hard | SL/HL | 25 marks

Topic 1.1: Modelling the economy: leakages and injections; size of the flow

(a) [10] Explain how an excess of planned leakages over planned injections can cause the circular flow of income to contract.

(b) [15] Using real-world examples, discuss whether reducing leakages is preferable to increasing injections when an economy's income flow is contracting.

Part (a) - Indicative answer and marking guidance

Definitions. Leakages are saving, taxation and import expenditure. Injections are investment, government purchases and export expenditure. Planned flows refer to intended spending and withdrawals; the model describes adjustment, not an accounting impossibility.

Reasoning. If intended withdrawals exceed intended additions at the current income, spending on domestic production is not fully replaced. Firms may face reduced sales or unwanted stocks, lower production and factor payments, and reduce household income. Explain how this feeds into a smaller flow; no numerical multiplier is required.

Diagram. No diagram is compulsory. If used, a five-sector circular flow should show S, T and M leaving and I, G and X entering the main loop. Mark the intended net withdrawal or use a labelled sequence of lower spending, output and income. A flow diagram has no required price axis or welfare area.

Common errors. Requiring S=I, T=G and M=X separately for overall balance; claiming ex-post accounting identities cease to hold; treating imports as wasted resources; saying the whole flow stops instantly.

Alternatives. Credit the equivalent statement S+T+M > I+G+X for planned flows, with a developed adjustment explanation. Do not require national-accounts identities or unplanned-investment terminology for full credit.

Part (b) - Indicative answer and marking guidance

Main argument. Lower planned withdrawals can preserve spending on domestic output. Higher injections can replace lost expenditure and may add useful production or capacity. Each route can address a net spending shortfall within the model.

Counterargument and conditions. Saving supports security and possible investment, taxes support public activity, and imports can supply useful inputs. Reducing one leakage may also reduce an injection or impair future options. Raising injections also uses resources and may generate offsetting leakages.

Real-world application. S4: pandemic saving and reduced spending provide a leakage-side example. S5: reconstruction provides an injection-side example. Treat these as distinct contexts, not a controlled experiment proving which route is superior.

High-quality evaluation and judgment. A strong judgment compares the source of the contraction, linked flows, feasibility and future resource consequences. Detailed tax instruments, monetary-policy channels or fiscal multipliers are not needed.

Question 19

Hard | SL/HL | 25 marks

Topic 1.1: Economics as a social science: microeconomics and macroeconomics; interdependence

(a) [10] Explain how a decision by one household to reduce consumption can differ in its effects from a similar decision by many households.

(b) [15] Using real-world examples, evaluate the view that choices which improve individual households' economic well-being necessarily improve economic well-being for society as a whole.

Part (a) - Indicative answer and marking guidance

Definitions. Microeconomics examines individual decision-makers and markets; macroeconomics examines the economy as a whole. Consumption is household expenditure on goods and services. Interdependence links one person's expenditure to another's income.

Reasoning. One household may retain income and improve its financial position with little economy-wide effect. Widespread reductions in consumption can reduce firms' receipts and factor incomes if other spending does not offset them. Explain the circular link and the distinction in scale; avoid assuming every household's original income stays fixed.

Diagram. No diagram required. If used, a household-firm loop should identify reduced C, then reduced firm receipts and factor income. S may be shown as a withdrawal. A qualitative circular-flow diagram is sufficient; neither AD-AS nor a multiplier diagram is required.

Common errors. Treating aggregate outcomes as a simple sum with every other variable unchanged; asserting that reduced consumption always lowers total income regardless of other flows; treating private well-being as only current spending.

Alternatives. Credit reduced household financial vulnerability and increased saving as the individual motive. Other offsetting injections or productive uses of saving are valid qualifications.

Part (b) - Indicative answer and marking guidance

Main argument. Choices that improve security or future consumption can improve household well-being and may support useful investment or reduce resource pressure. Private and wider gains can coincide.

Counterargument and conditions. Other households and firms may experience reduced income when spending falls collectively. Households have different starting circumstances and can bear different consequences. The composition of spending and offsetting injections affect the wider result.

Real-world application. S4: the Bank of England found uneven saving gains during the pandemic alongside weaker spending. Explain how the same episode can include greater security for some and income losses for others; do not claim the source establishes a universal causal law.

High-quality evaluation and judgment. A strong judgment explains when individual and collective interests align and when interdependence creates a divergence. Topic 2.8 externality diagrams and formal paradox-of-thrift or multiplier models are not required.

Question 20

Medium | SL/HL | 25 marks

Topic 1.1: The problem of choice: land, labour, capital and entrepreneurship; PPC

(a) [10] Explain how entrepreneurship combines the other factors of production to produce goods and services.

(b) [15] Using real-world examples, discuss the view that natural resource endowments are the main determinant of an economy's production possibilities.

Part (a) - Indicative answer and marking guidance

Definitions. Land comprises natural resources; labour is human effort; capital is produced productive equipment and structures. Entrepreneurship organises production, identifies opportunities and bears uncertainty associated with resource commitments.

Reasoning. Explain how an entrepreneur selects an output, assembles an appropriate combination of natural inputs, workers and capital, and coordinates their use. Connect decisions about what and how to produce to scarce resources and forgone alternatives. Merely naming the four factors does not meet the full explanatory demand.

Diagram. No diagram required. A PPC may illustrate complementary improvements in productive resources, but no firm revenue, cost or profit-maximisation curves are expected.

Common errors. Treating entrepreneurship as money alone; describing capital as any natural asset; assuming an entrepreneur necessarily owns all inputs; relying on a one-sentence list of factor definitions.

Alternatives. Credit cooperative, public-enterprise or small-business organisation if the entrepreneurial function is clearly identified. Do not require every venture to make a profit or invent a new product.

Part (b) - Indicative answer and marking guidance

Main argument. Resource endowments constrain feasible production and can make some activities possible on a large scale. Scarce water, usable land or energy can limit an economy despite strengths elsewhere.

Counterargument and conditions. Resources must be combined with skills, equipment, organisation and technology. Resource-poor economies can expand capacity through these complementary factors; abundant natural inputs need not be used effectively. Importance depends on the output mix and resource bottlenecks.

Real-world application. S1: Singapore's water constraints and technological response demonstrate both the relevance and limits of natural endowments. S6: Korea's development through skills, infrastructure and technology offers a complementary application without assuming resource abundance.

High-quality evaluation and judgment. A strong answer makes a comparative judgment about natural inputs and the other determinants of productive capacity. It should avoid claiming either that endowments determine everything or that technology makes them irrelevant.

Case notes and source register

These concise notes establish factual anchors; the question-specific economic interpretations are applications of theory. Suggested examples are not mandatory. Sources were checked on 19 October 2026.

G1. International Baccalaureate, Economics guide, first assessment 2022. Topic 1.1: printed pp. 22-23; Paper 1 structure: pp. 60-62; markbands: pp. 63-65. Official publication hosted by an IB teaching site.

Open source

S1. PUB, Singapore Water Story. Water scarcity motivated investment in recycled water and desalination alongside catchment and imported supplies. This supports analysis of technology, scarce inputs and public allocation.

Open source

S2. UK Government, The NHS Constitution for England. The NHS links access to clinical need rather than ability to pay, while committing to effective, fair use of finite resources. This illustrates access choices and opportunity cost.

Open source

S3. World Bank, Trade Reform in Vietnam: Opportunities with Emerging Challenges (2003). The report describes the transition initiated in 1986 from central planning towards a market-oriented system with a continuing government role.

Open source

S4. Bank of England, How has Covid affected household savings? (25 November 2020). The survey found reduced spending and increased saving for some households, with gains unevenly distributed. This supports conditional circular-flow analysis, not a universal causal estimate.

Open source

S5. World Bank, Economic impacts (2012), Great East Japan Earthquake knowledge note. Production disruption and reconstruction provide a setting for distinguishing damage to productive assets, utilisation and recovery.

Open source

S6. World Bank, Republic of Korea country overview. It identifies sustained investment in education, infrastructure and technology as part of Korea's transformation. This supports analysis of complementary inputs and expanding capacity.

Open source

S7. World Bank, World Bank Report Outlines Path to High Income for Viet Nam Through Higher Value Trade and Economic Reforms (21 November 2024). It describes export-related activity and the importance of stronger domestic production linkages.

Open source

Uploaded reference materials

IB Economics SL and HL subject briefs (first assessments 2022); Ellie Tragakes, Economics for the IB Diploma, second edition, Cambridge University Press, 2012; supplied master prompt and six question screenshots. The conceptual checks and originality review use the source hierarchy in the master prompt.

Publication check

20 matched pairs; 10/15 marks throughout; all SL/HL; no student-facing answers or difficulty labels; required diagrams specified in the teacher entries; 4 Easy / 10 Medium / 6 Hard. Each pair has a different primary assessment focus. Related core concepts recur where needed for explanation. Topic 1.1's AO2/AO4 scope and the adapted nature of part (b) are disclosed at the start of this guide.